Applications of integration in life science

BMB547

Topics today

- **Hagen–Poiseuille Law: Liquid flow through a pipe / artery**
- **Modelling Tumor Growth using the Gompertz Function**

After this lecture, you should be able to:

•Explain **why calculus (integration + differential equations)** is useful for **building quantitative models** of biological and biomedical systems.

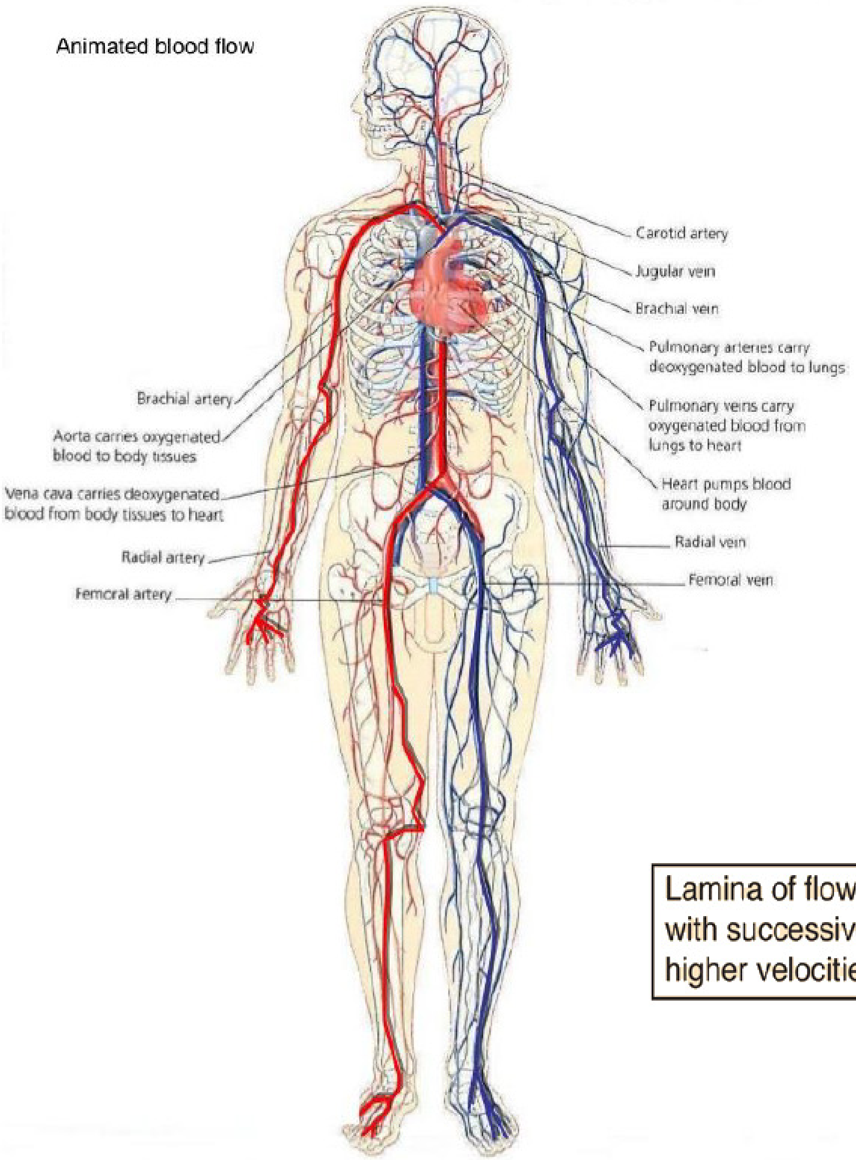
•Use **integration** to derive a measurable prediction (e.g. **flow rate in a blood vessel** from a velocity profile).

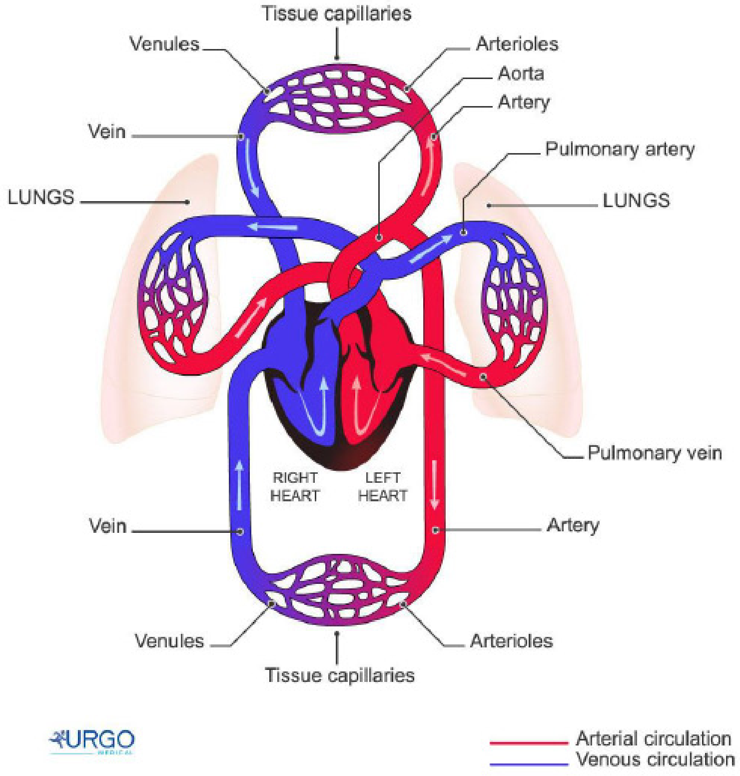
•Interpret what the math means biologically (e.g. why small changes in **vessel radius** strongly affect blood flow).

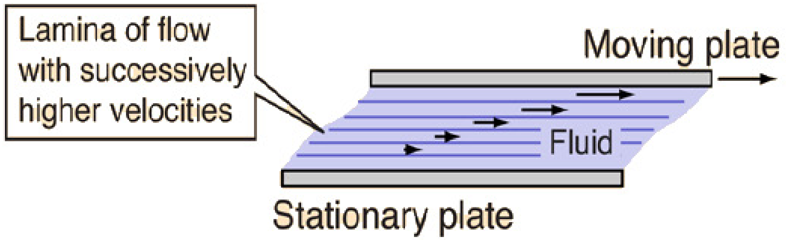
•Set up a **model from a biological question** by defining variables, assumptions, and what is being predicted.

•Solve and interpret a simple **growth model** (Gompertz) and connect parameters to tumour growth behaviour.

Blood flow in a capillary







Research Paper

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Elastin Organization in Pig and Cardiovascular Disease Patients' Pericardial Resistance Arteries

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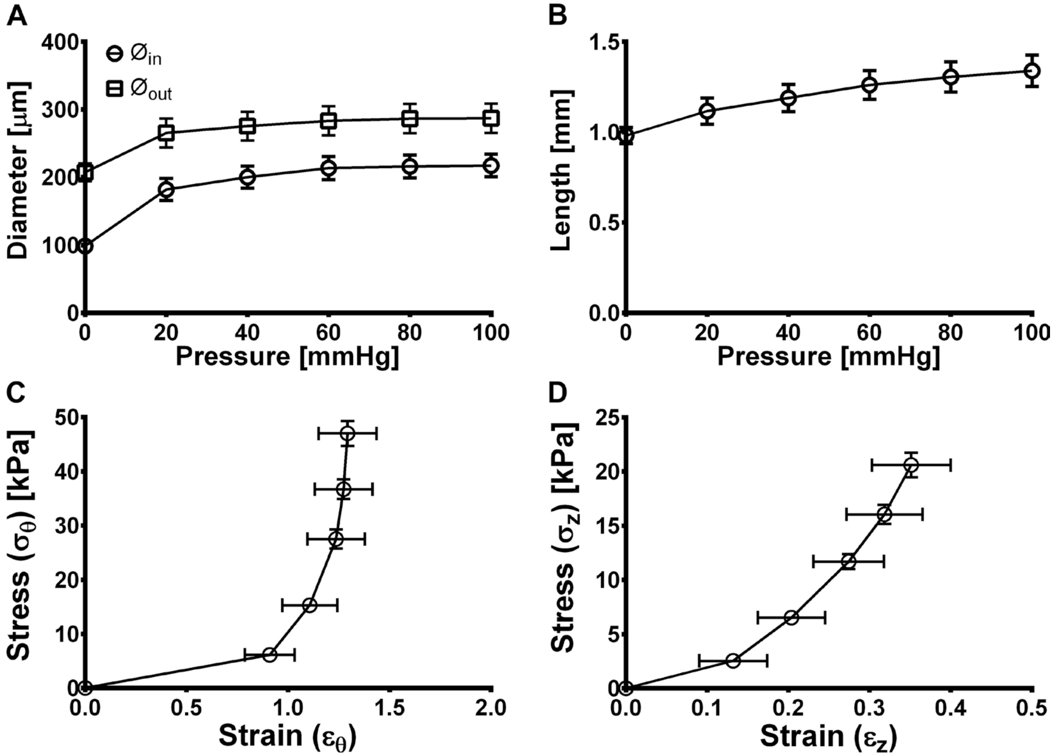
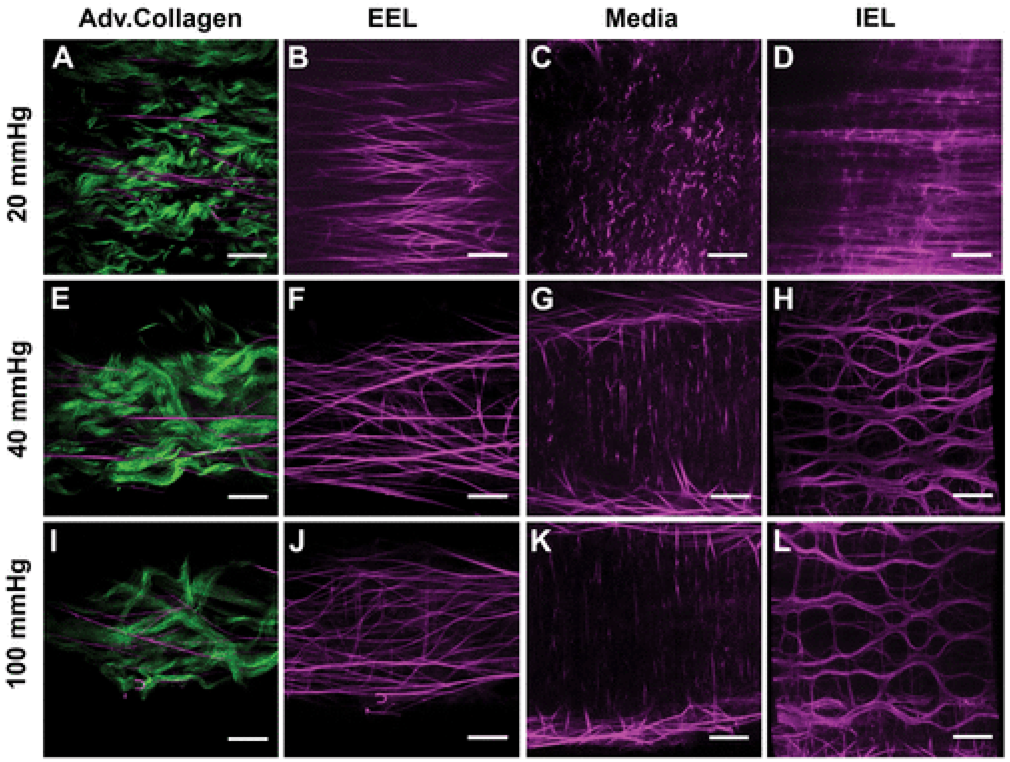
Imaging and modeling of acute pressure-induced changes of collagen and elastin microarchitectures in pig and human resistance arteries

- Peripheral vascular resistance is increased in essential **hypertension**.
- The pathophysiology behind this involves structural remodeling of resistance-sized arteries (Ø100-300 µm) and stiffening of their arterial wall.
- The molecular basis of this remodeling has been studied in experimental animal models and human patients.

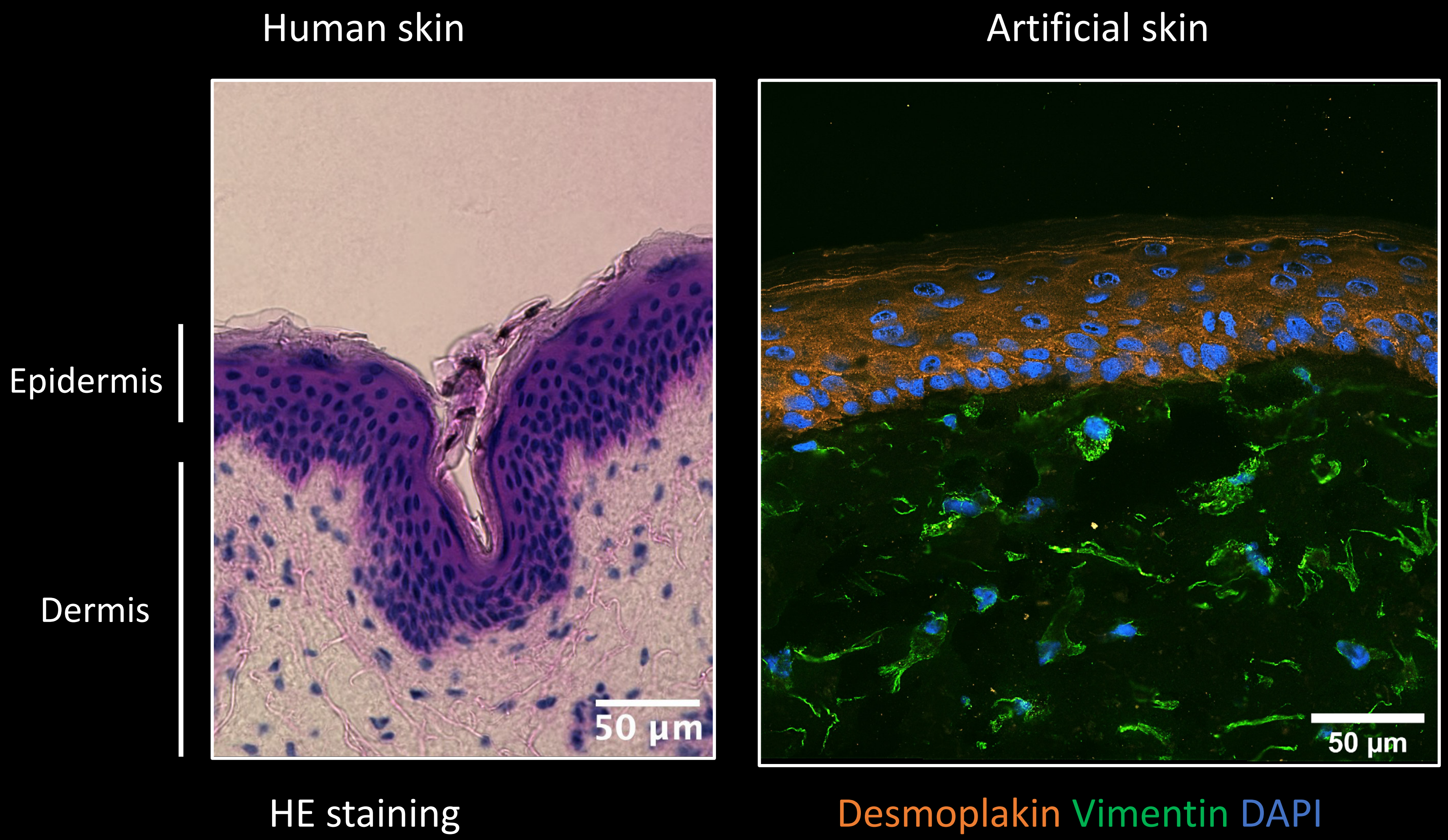


Structural and mechanical characteristics of pig pericardial resistance arteries (pPRAs) as a function of increasing transmural pressure. *A*.



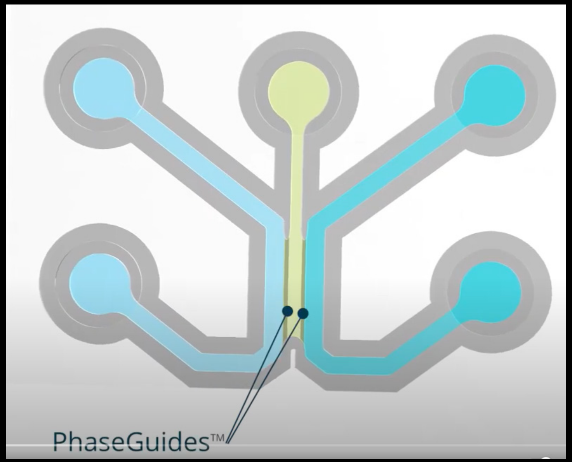
Two-photon excitation fluorescence microscopy images of collagen (green) and elastin (magenta).Scale bars = 10 µm.

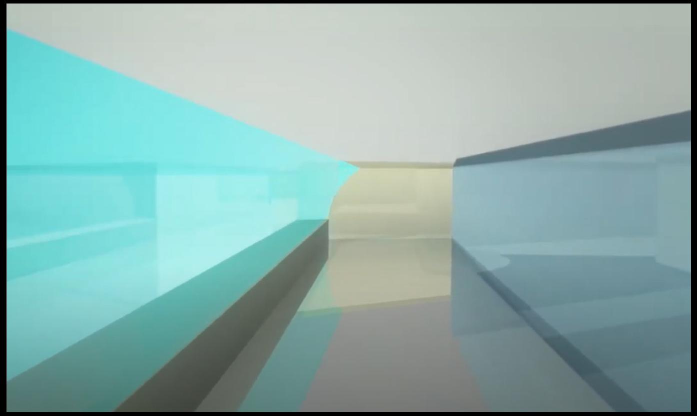
Artificial skin

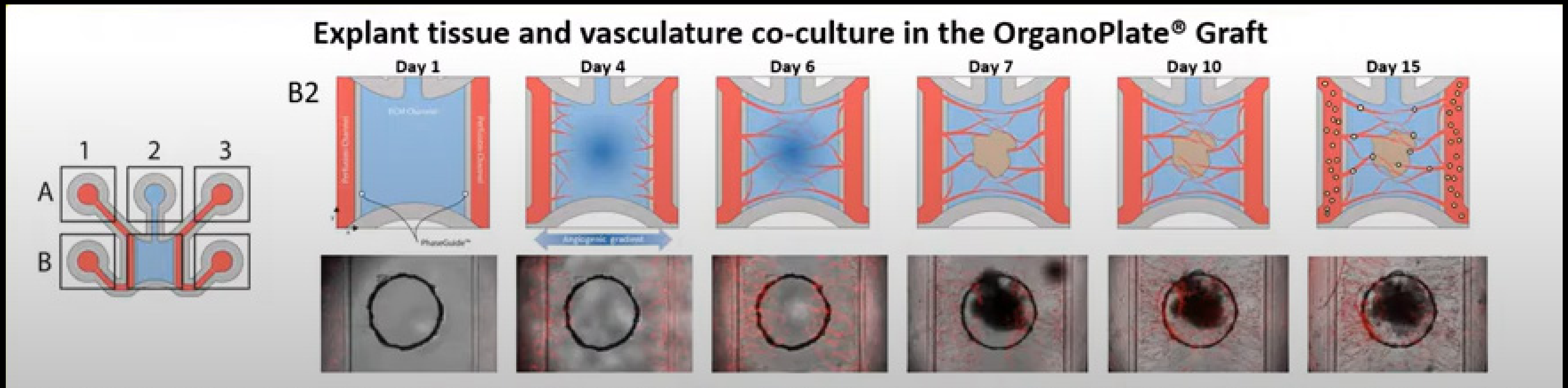


Organ on a chip (vascularization of artificial skin)

- Used for vascularization of tissues (here shown tumor)
- Well-plate format – glass bottom







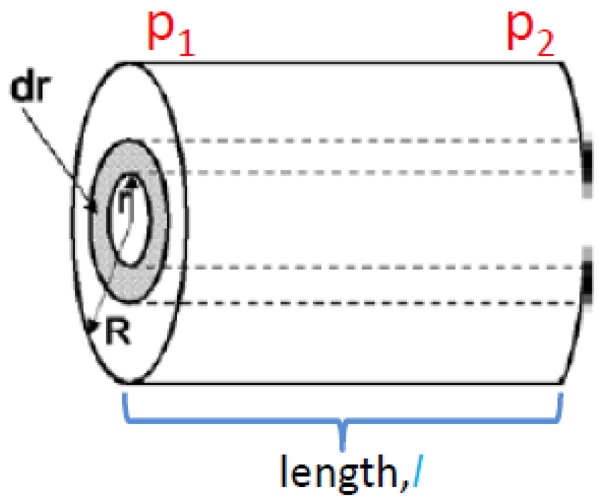
Establishment of the co-culture requires **1) blood vessel 3D tubes** formation via seeding of endothelial cells in the perfusion lanes; **2)** addition of an angiogenic cocktail[2] and subsequent **vascular bed development;** and **3) placement of tumor explant** on the central chamber.

Blood flow in a capillary

Find the velocity

Laminar flow has two forces:

1. Due to friction $F_r = -\eta A_c \frac{dv}{dr}$ $F_r = -\eta 2\pi r l \frac{dv}{dr}$



force balance → 2) separate variables → 3) integrate → 4) apply boundary → 5) integrate area to get $Flux$.

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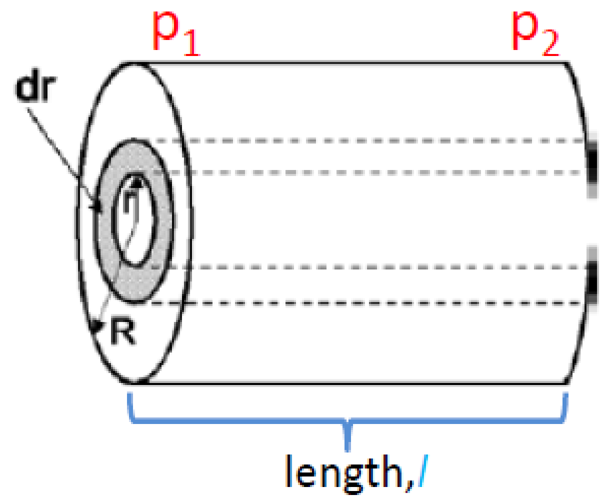

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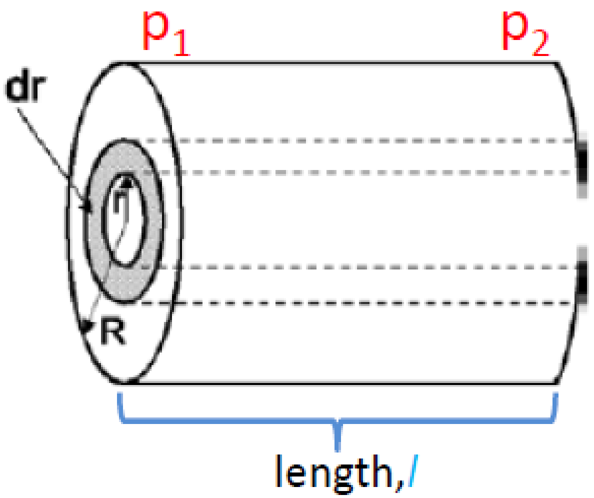
Forces balance each other:

$$F_r + F_p = 0 \Rightarrow F_p = -F_r$$

$$\Delta p \pi r^2 = \eta 2\pi r l \frac{dv}{dr}$$

Seperate the variables:

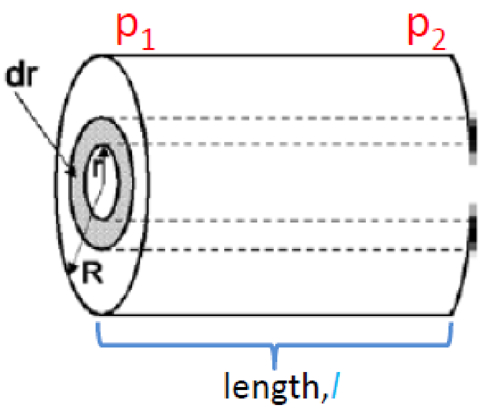
With that: $dv = \frac{\Delta p \pi r^2}{\eta 2\pi r l} dr$ ...



force balance → 2) **separate variables** → 3) integrate → 4) apply boundary → 5) integrate area to get $Flux$.

Blood flow in a capillary

With that: $dv = \frac{\Delta p \pi r^2}{\eta 2 \pi r l} dr$ ...upon integration over the radius of the capillary



$$v(r) = \int_r^R \frac{1}{2\eta} \frac{\Delta p}{l} r dr$$

force balance → 2) separate variables → 3) **integrate** → 4) apply boundary → 5) integrate area to get $Flux$.

Blood flow in a capillary

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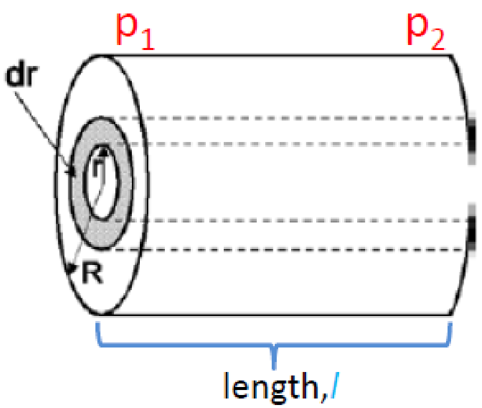

$$v(r) = \int_r^R \frac{1}{2\eta}\frac{\Delta p}{l} r\,dr = \frac{1}{2\eta}\frac{\Delta p}{l}\int_r^R r\,dr$$

force balance → 2) separate variables → 3) **integrate** → 4) apply boundary → 5) integrate area to get flux.

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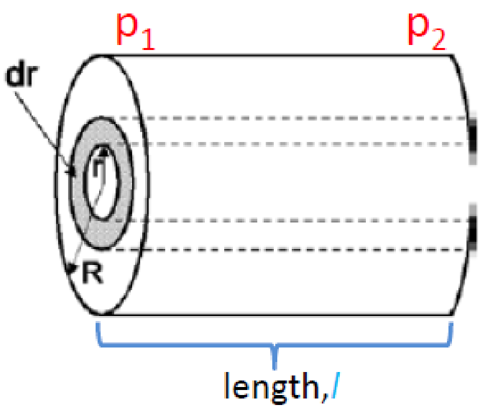
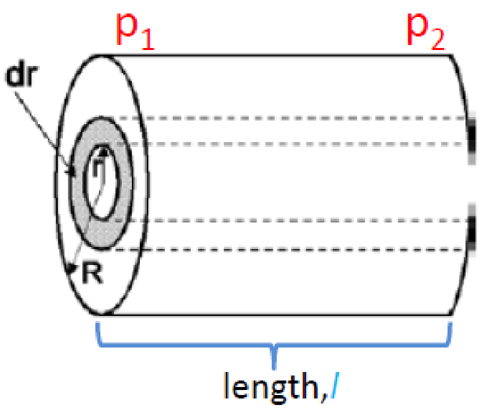


$$v(r) = \int_r^R \frac{1}{2\eta}\frac{\Delta p}{l} r dr = \frac{1}{2\eta}\frac{\Delta p}{l}\int_r^R r dr = \frac{1}{2\eta}\frac{\Delta p}{l}\frac{r^2}{2}\Big|_r^R + C$$

force balance → 2) separate variables → 3) **integrate** → 4) apply boundary → 5) integrate area to get flux.

Blood flow in a capillary

With that: $dv = \frac{\Delta p \pi r^2}{\eta 2\pi r l} dr$ ...upon integration over the radius of the capillary



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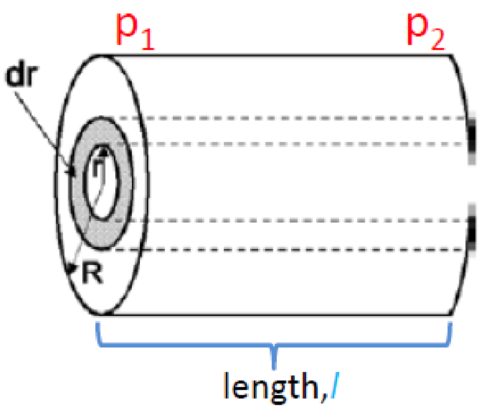
Boundary conditions: v(r=R) = 0

force balance → 2) separate variables → 3) integrate → 4) **apply boundary** → 5) integrate area to get flux.

Blood flow in a capillary

With that: $dv = \frac{\Delta p \pi r^2}{\eta 2\pi r l} dr$ ...upon integration over the radius of the capillary



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with v(r=R)=0 is C=0. Thus, we get for the velocity profile:

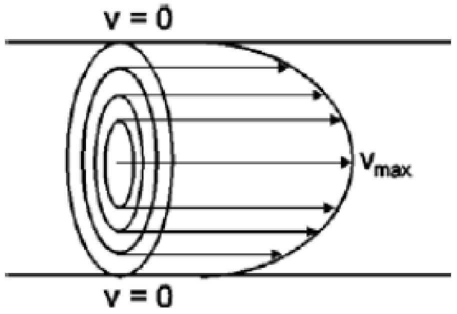
$$\boxed{v(r) = \frac{1}{4\eta}\boxed{\frac{\Delta p}{l}}(R^2 - r^2)}$$

force balance → 2) separate variables → 3) integrate → 4) **apply boundary** → 5) integrate area to get flux.

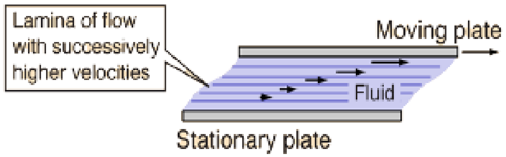
Blood flow in a capillary

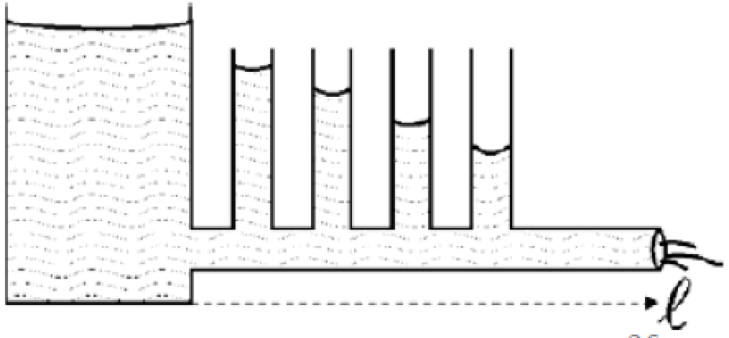
$$v(r) = \frac{1}{4\eta}\boxed{\frac{\Delta p}{l}}(R^2 - r^2)$$

The longer the pipe
The lower the velocity

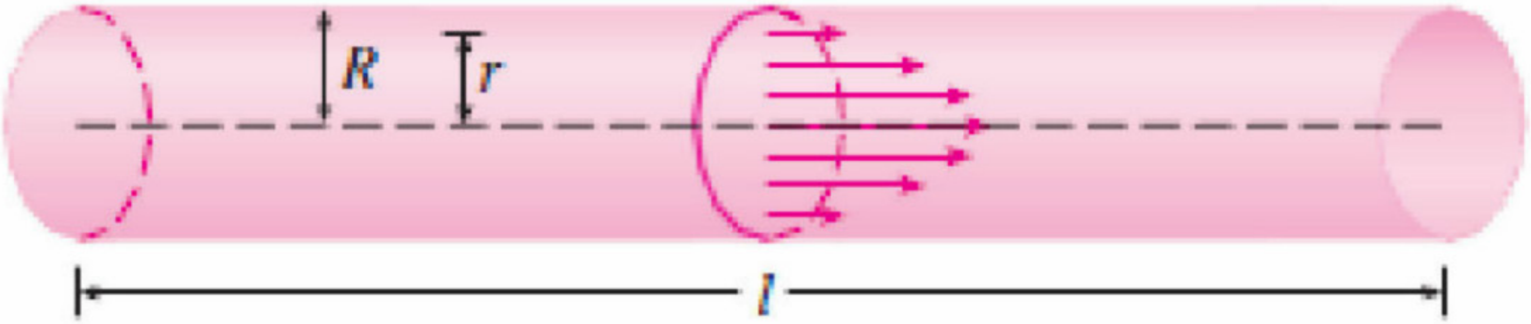


The flow speed is highest in the middle of the capillary.





Now let's find the rate of blood flow



We can model the shape of a blood vessel (artery or vein) using a cylindrical tube. Due to the friction caused by the walls, the velocity v of the blood is greatest along the central axis. It decreases as the radius r increases and reaches the wall of the artery and becomes 0. The relationship between v and r is given by the law of laminar flow which is:

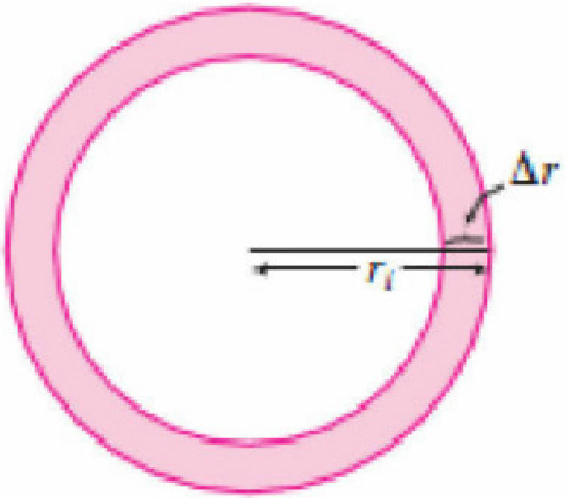
$$v = \frac{P}{4\eta l}(R^2 - r^2)$$

where η is the viscosity of the blood , P is the pressure difference between the ends of the tube and l is the length. If P and l are constant, then v is a function of r with **domain [0,R]**.

This equation gives us the velocity of blood that flows along the blood vessel with a radius R and length l at a distance r from the central axis.

To find the rate of the blood flow, or flux, we use smaller, equally spaced radii so that the ring has an inner radius r_{i-1} and outer radius r_i and the area of the ring is:

$2\pi r_i \Delta r$ where $\Delta r = r_i - r_{i-1}$



force balance → 2) separate variables → 3) integrate → 4) apply boundary → 5) **integrate area to get $flux$.**

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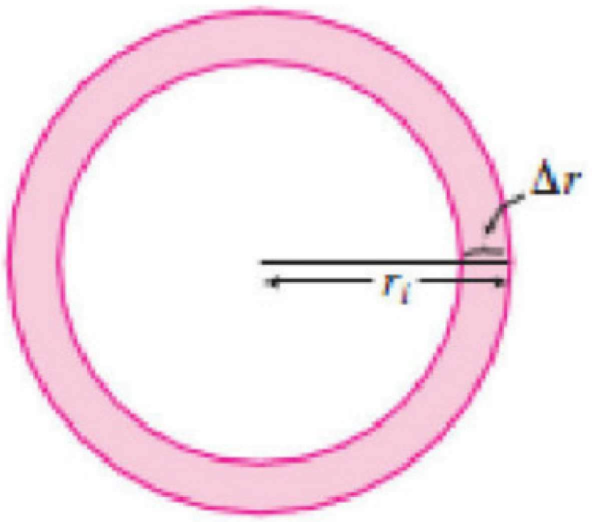
$2\pi r_i \Delta r$ where $\Delta r = r_i - r_{i-1}$

If Δr is small, the velocity is basically constant throughout the ring and is approximated by $v(r_i)$. The volume of blood per unit time is:

$2\pi r_i \Delta r\ v(r_i) = 2\pi r_i v(r_i)\Delta r$



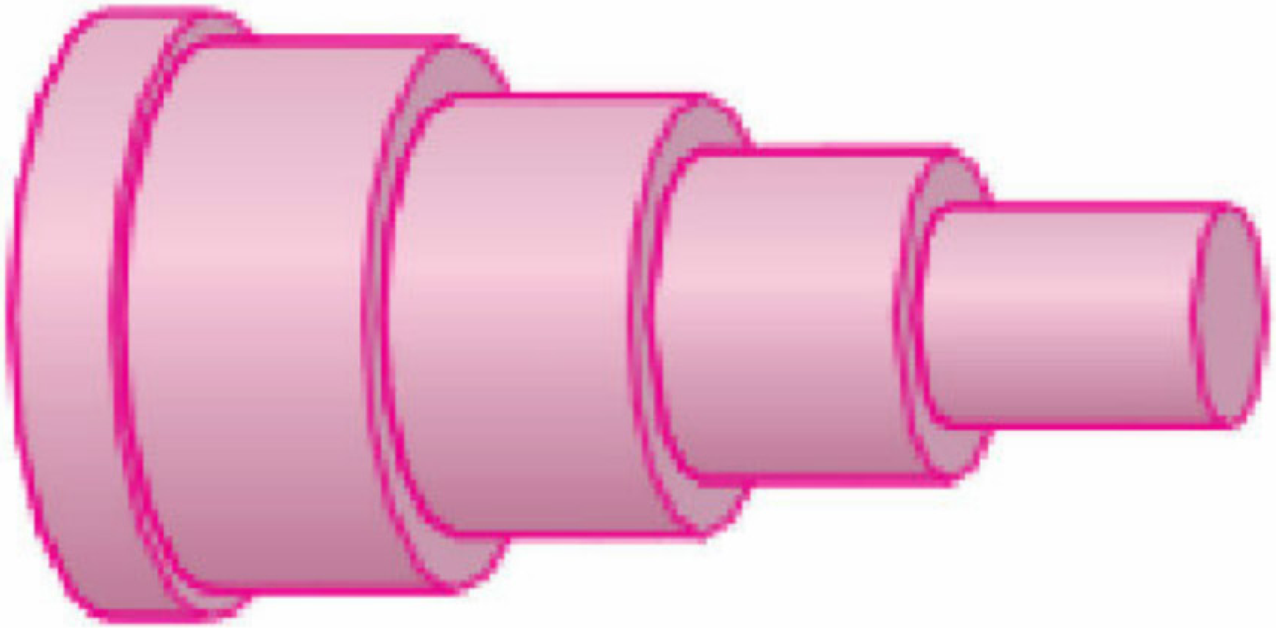
Are the units correct?

Area [m^2] x velocity [m/s] = [m^3/s]

force balance → 2) separate variables → 3) integrate → 4) apply boundary → 5) **integrate area to get $flux$.**

Therefore, the total volume of blood flowing trough the blood vessel is:

$$\sum_{i=1}^{n} 2\pi r_i v(r_i)\Delta r$$



force balance → 2) separate variables → 3) integrate → 4) apply boundary → 5) **integrate area to get $flux$.**

The velocity of blood increases in the centre of the blood vessel. The limit gives the exact value of the flux (volume of blood that passes through the blood vessel) per unit time.

$$F = \lim_{n\to\infty} \sum_{i=1}^{n} 2\pi r_i v(r_i)\Delta r = \int_0^R 2\pi r v(r)\, dr$$

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$$= \frac{\pi P}{2\eta l}\int_0^R (R^2 r - r^3)dr = \frac{\pi P}{2\eta l}\left[R^2\frac{r^2}{2} - \frac{r^4}{4}\right]_{r=0}^{r=R}$$

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$$= \frac{\pi P}{2\eta l}\left[\frac{R^4}{2} - \frac{R^4}{4}\right] = \frac{\pi P R^4}{8\eta l}$$

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$$= \frac{\pi P}{2\eta l}\left[\frac{R^4}{2} - \frac{R^4}{4}\right] = \frac{\pi P R^4}{8\eta l}$$

The final equation is:

$$F = \frac{\pi P R^4}{8\eta l}$$

which is known as the Poiseuille's Law. It shows that the flux is proportional to the fourth power of the radius of the blood vessel.

force balance → 2) separate variables → 3) integrate → 4) apply boundary → 5) **integrate area to get $flux$.**

Example:

Use Poiseuille's Law to calculate the rate of flow in a small human artery where we can take $\eta = 0.027, R = 0.008\ cm, l = 2cm$ and $P = 4000 dynes/cm^2$.

Solution:

$$F = \frac{\pi P R^4}{8\eta l}$$

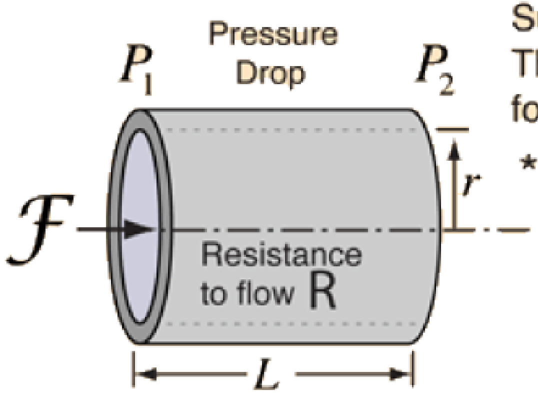
$$F = \frac{\pi\left(4000\frac{dynes}{cm^2}\right)(0.008cm)^4}{8(0.027)(2\ cm)}$$

$$F = 0.000119147$$

$$F = 1.19 \times 10^{-4}\ cm^3/s$$

$\therefore$ The flux is $1.19 \times 10^{-4}\ cm^3/s$.

Blood flow in a capillary



Suppose the original flowrate is 100 cm^3/sec. The effect of changes in the parameters is as follows:

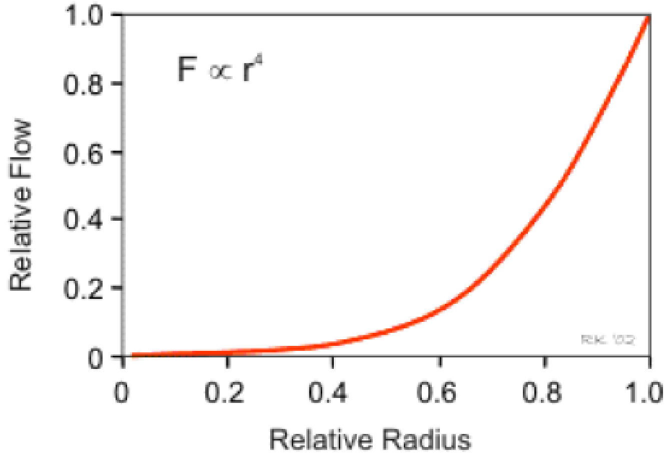
*Double length	⇨	50	cm^3/sec
Double viscosity	⇨	50	cm^3/sec
Double pressure	⇨	200	cm^3/sec
Double radius	➡	1600	cm^3/sec

* With other parameters held at original values

$$\mathcal{R} = \frac{8\eta L}{\pi r^4}$$ where η = viscosity

$$\text{Volume Flowrate} = \mathcal{F} = \frac{P_1 - P_2}{\mathcal{R}} = \frac{\pi(\text{Pressure difference})(\text{radius})^4}{8(\text{viscosity})(\text{length})}$$

A 19% increase in radius will double the volume flowrate!



Hagen-Poiseuille Law

$$\boxed{i = \frac{dV}{dt} = \frac{\pi}{8\eta}\frac{\Delta p}{l}R^4}$$

Implications of Hagen Poisseulle's Law in Biomedicine?

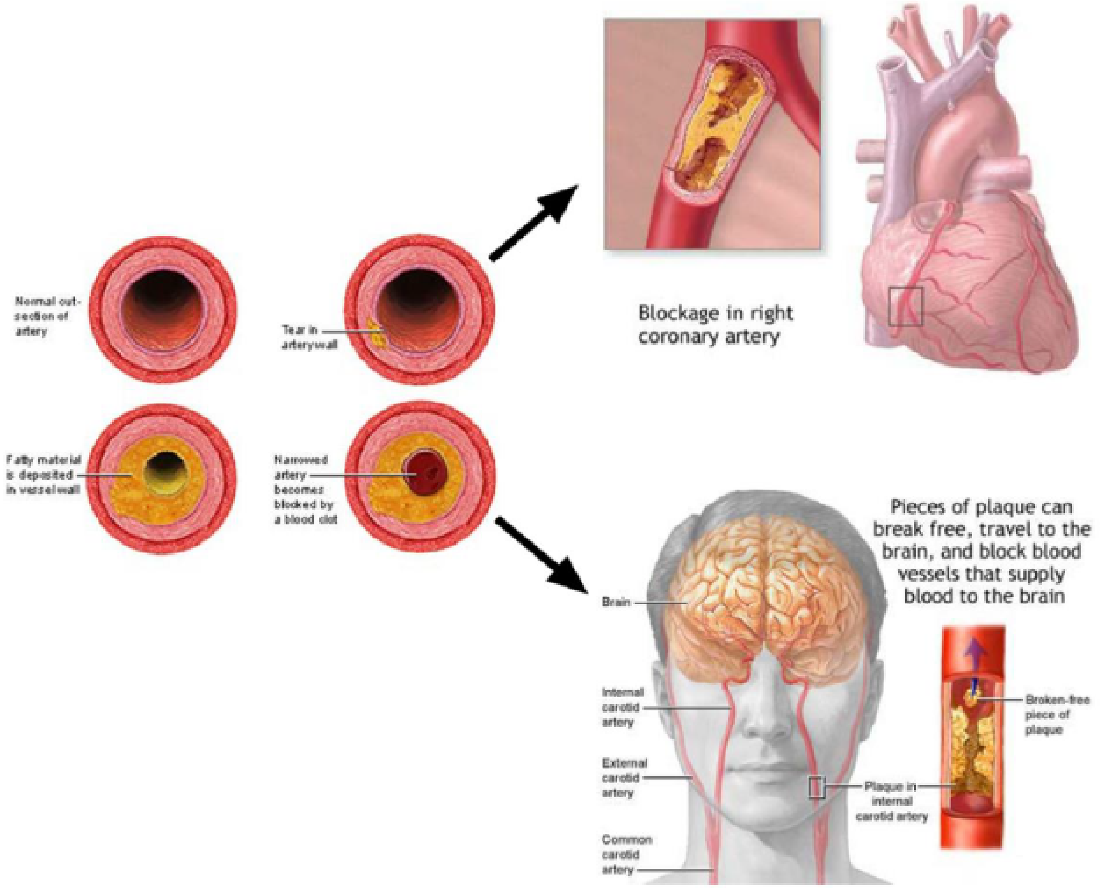
Hagen-Poiseuille Law

$$\boxed{i = \frac{dV}{dt} = \frac{\pi}{8\eta}\frac{\Delta p}{l}R^4}$$

Blood flow in a capillary

Implications of Hagen-Poisseuille's Law in Biomedicine:

- Constricting the width by factor 2 would decrease blood flow 16fold! To avoid that, the heart pumps more, but this causes much higher pressure in the artery:



Consequences:

To keep same flow rate, the heart must pump much more (16fold pressure)!

The higher pressure causes higher forces, due to: $F_p = (p_2 - p_1) \cdot A = \Delta p \pi r^2$

This gives higher friction, due to:

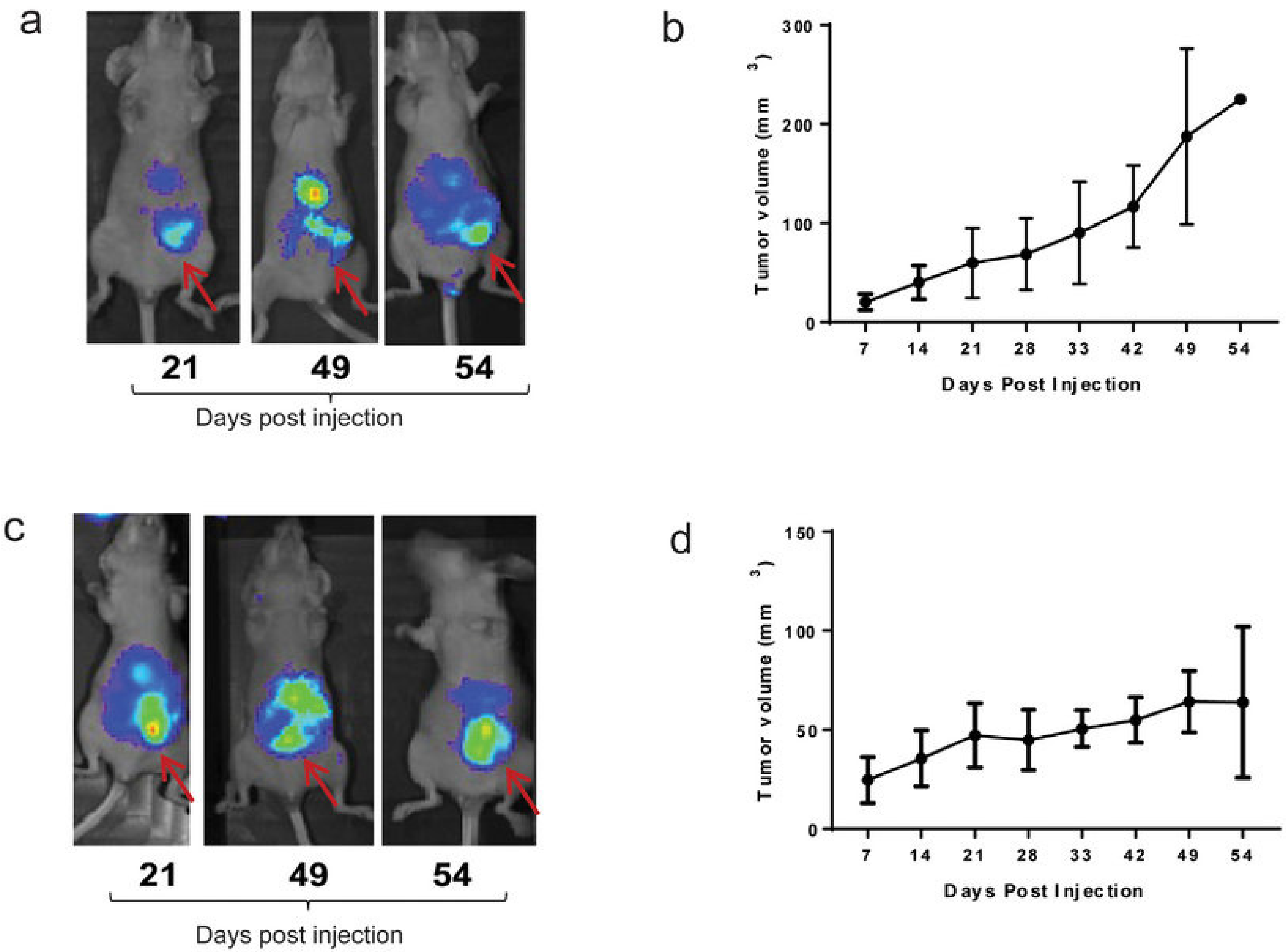
$$\Delta p \pi r^2 = \eta 2 \pi r l \frac{dv}{dr}$$

This increases the risk, that small pieces of the atherosclerotic plaque break of evtl. causing stroke!

Hagen-Poiseuille Law

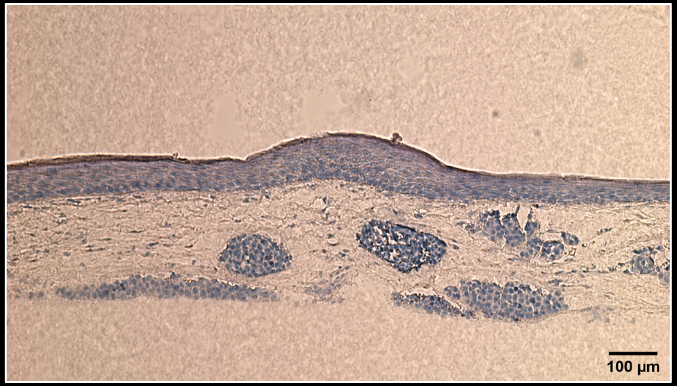
$$\boxed{i = \frac{dV}{dt} = \frac{\pi}{8\eta} \frac{\Delta p}{l} R^4}$$

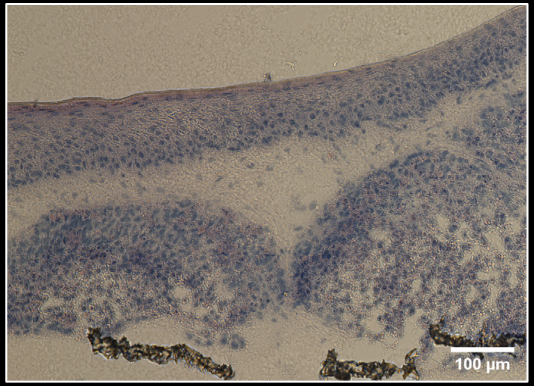
Modelling Tumor Growth using the Gompertz Function



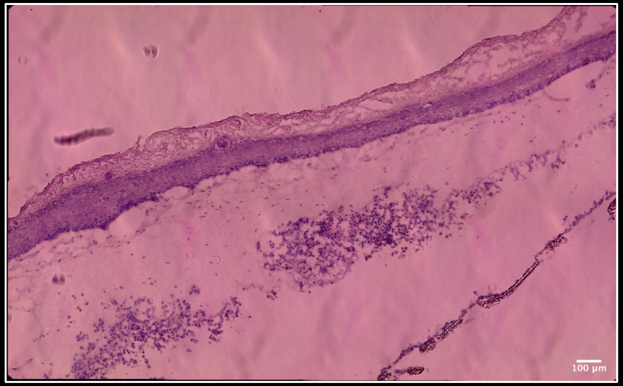
Skin cancer model

A375 spheroids in full thickness model



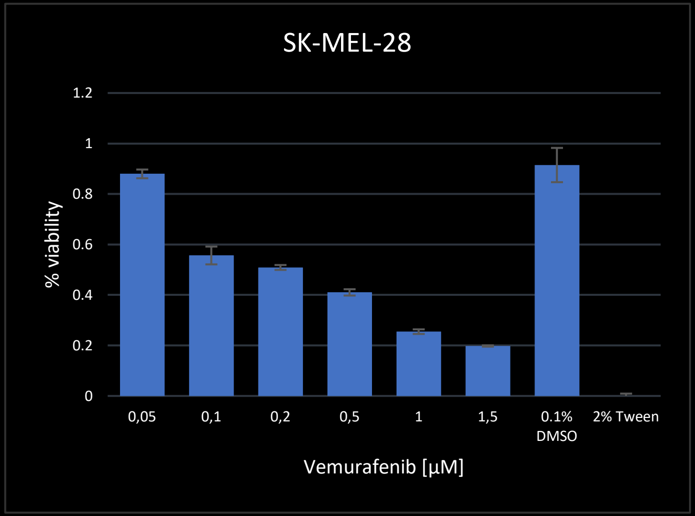


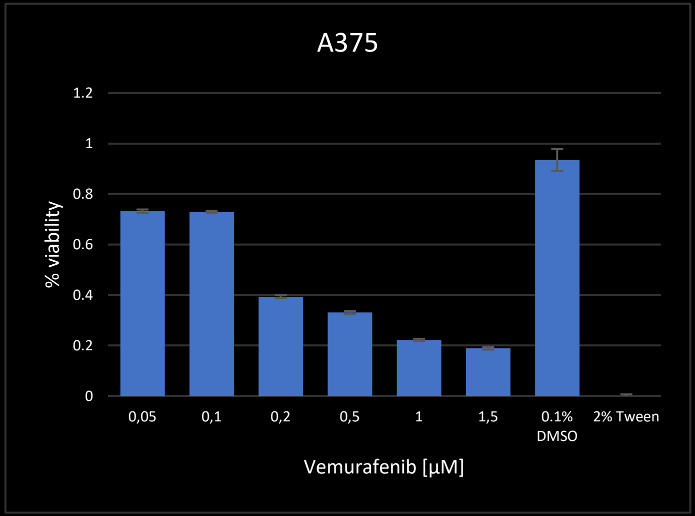
SK-MEL-28 spheroids in full thickness model



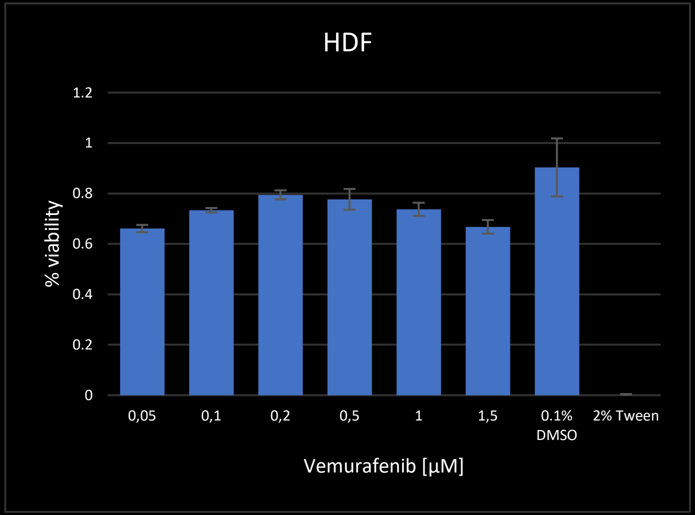
HE staining

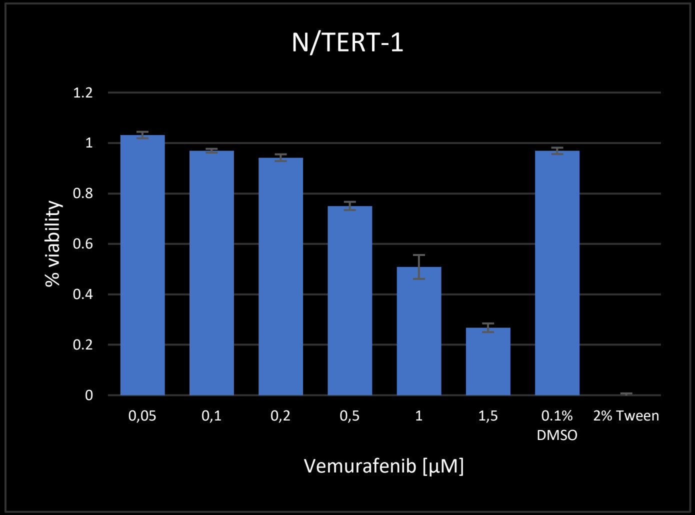
Cytotoxicity assay – vemurafenib



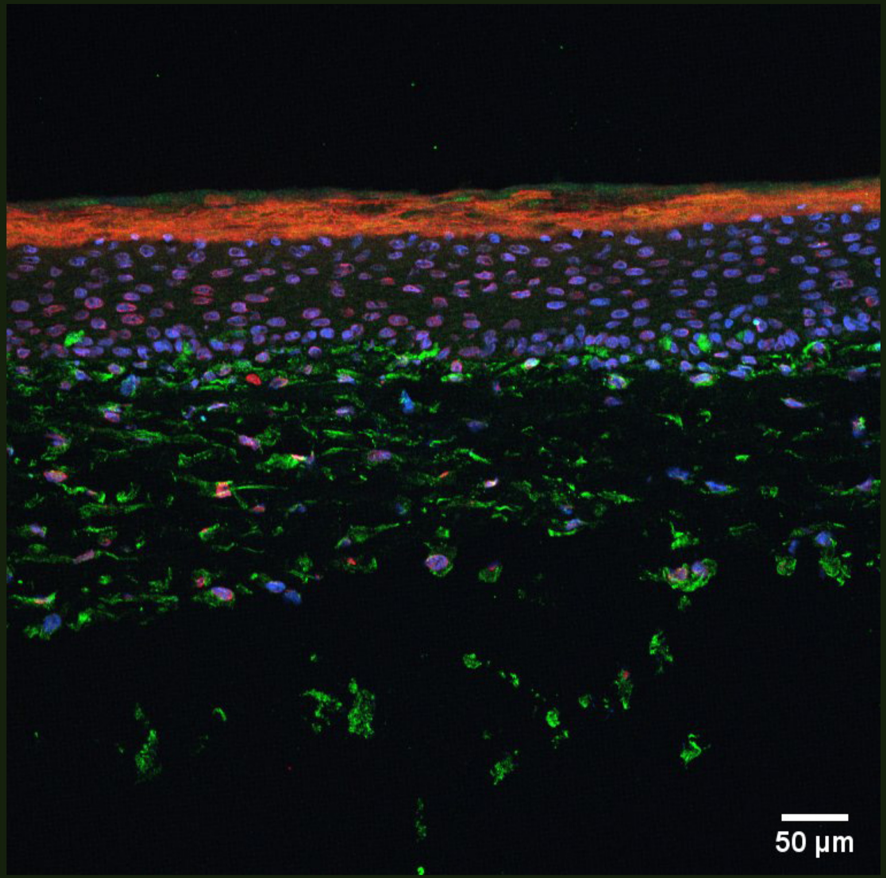
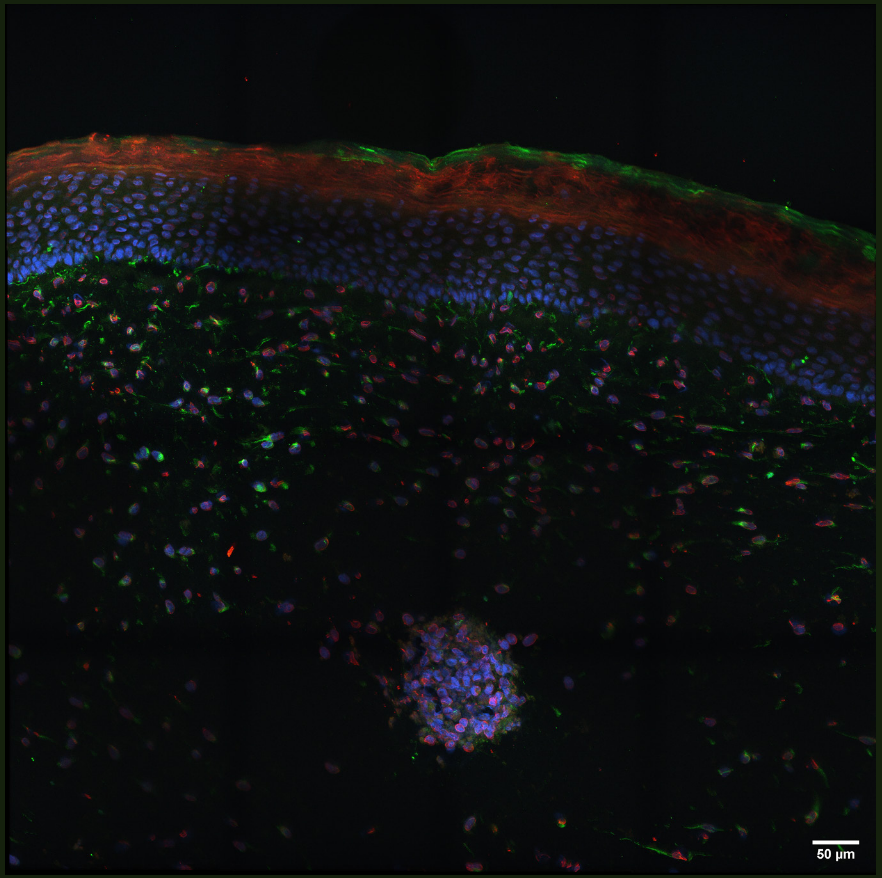


- 72 hours treatment
- Fluorescent intensity read on a plate reader after incubation with CellTiter-Blue

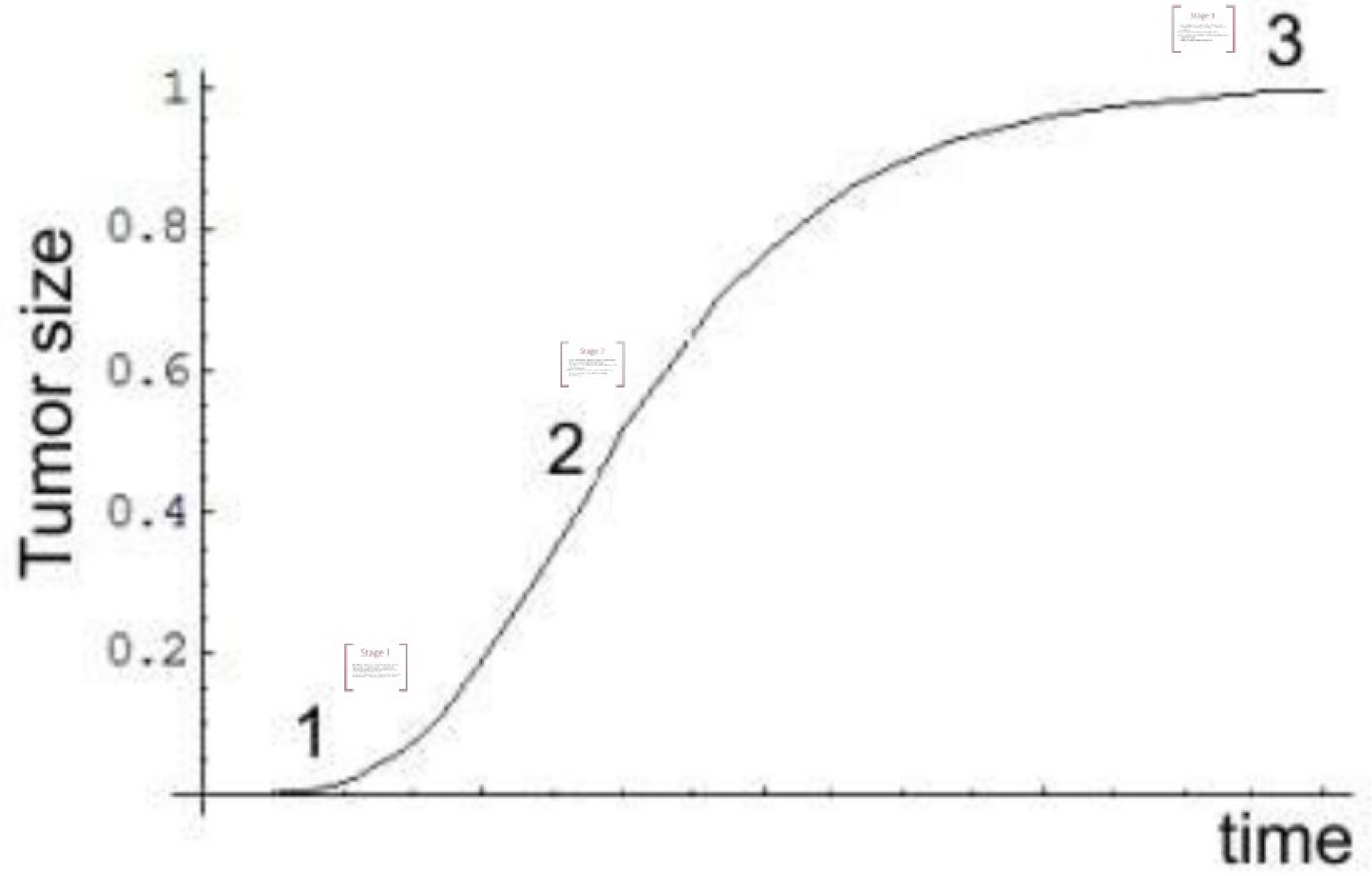




Ki67 (red) + S100B (green) in skin models

No melanoma tumors	A375 melanoma tumors
	

Relation to Tumor Growth



Stage 1

- Normal (non-cancerous) tissues are capable of growth
- Homeostatic regulatory system is responsible for equilibrium between cell multiplication and cell loss
- Cancer entails a failure of the HRS.
- Abnormal multiplication in cancerous tissue then makes the cell multiplication/loss equilibrium unattainable

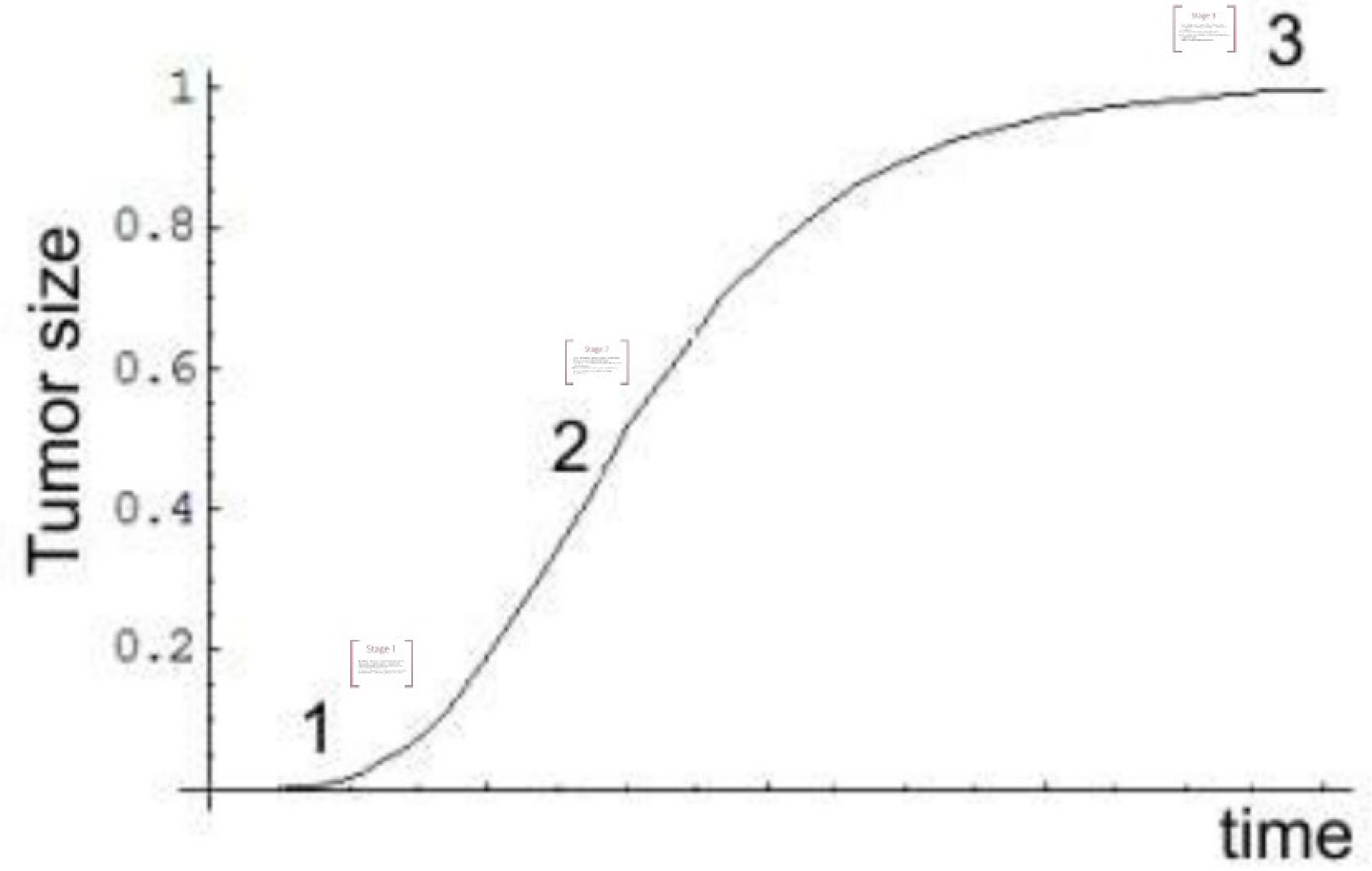
Stage 2

- As cell population rapidly increases, neighbouring blood vessels are pulled into the tissue.
- Cells begin to formulate new blood vessels, a process called Angiogenisis
- Now a tumour, it is connected to the circulatory system – subsequent growth is very large (exponential)

Stage 3

- Such rapid growth means that the tumour soon outstrips its blood supply, leading to a decrease in growth rate
- It will eventually stop growing altogether
- This implies that the size of the cell population has a asymptotic limit
- More of a mathematical abstraction

Relation to Tumor Growth



Need to make a model

$$\frac{dV}{dt} = V * a \Rightarrow \ exponetial\ growth$$

$$\frac{dV}{dt} = V * (a - b * \ln(V))$$

V = tumor volume
a = growth constant
b = is a parameter that measures how rapidly the curve departs from the simple exponential shape into a sigmoid shape

Find solution for Gompertz equation where V(t=0)=V_0

$$\frac{dV}{V * (a - b * \ln(V))} = 1 * dt$$

How to solve this?   Substitution

model → 2) Rewrite ODE → 3) Substitute → 4) Integrate → 5) Apply boundary V_0

Remember for last lecture:

Substitution method

$$\int f(g(x))g'(x)\mathrm{d}x = F(g(x)) + c$$

With the auxiliary variable $u = g(x)$, it follows that $\frac{\mathrm{d}u}{\mathrm{d}x} = g'(x)$, and we can write

$$\int f(u)\mathrm{d}u = F(g(x)) + c$$

$$\frac{dV}{V * (a - b * \ln(V))} = 1 * dt$$

$$\frac{1 * dV}{V * U} = 1 * dt$$

$Define$: $U = a - b * \ln(V)$

Then $\frac{dU}{dV} = 0 - b * \frac{1}{V} = -b * \frac{1}{V}$

And $dV * \frac{1}{V} = -\frac{1}{b}\, dU$

model → 2) **Rewrite ODE** → 3) Substitute → 4) Integrate → 5) Apply boundary V_0

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$$\frac{dV}{V * (a - b * \ln(V))} = 1 * dt$$

$$\frac{1 * dV}{V * U} = 1 * dt$$

$$-\frac{1}{b} * \frac{dU}{U} = 1 * dt$$

$Define: U = a - b * \ln(V)$

Then $\frac{dU}{dV} = 0 - b * \frac{1}{V} = -b * \frac{1}{V}$

And $dV * \frac{1}{V} = -\frac{1}{b}\, dU$

model → 2) Rewrite ODE → 3) **Substitute** → 4) Integrate → 5) Apply boundary V_0

Now integrate on both sides

$$-\frac{1}{b}\int \frac{dU}{U} = \int 1 * dt \qquad \rightarrow \qquad -\frac{1}{b}\ln(U) = t + C \;\; where\ C\ is\ a\ constant$$

model → 2) Rewrite ODE → 3) Substitute → 4) **Integrate** → 5) Apply boundary V_0

$$-\frac{1}{b}\ln(U) = t + C \qquad where\ C\ is\ a\ constant$$

$\Downarrow$

$$\ln(U) = -b * t + C_0$$

$\Downarrow$

$$U = C_0 \exp(-b * t) \qquad insert\ U = a\ - b * \ln(V)$$

$\Downarrow$

$$a\ - b * \ln(V) = C_0 \exp(-b * t)$$

$\Downarrow$

$$\ln(V) = \frac{a}{b} - \frac{C_0}{b}\exp(-b * t)$$

$\Downarrow$

$$V = \exp(\frac{a}{b} - \frac{C_0}{b}\exp(-b * t))$$

model → 2) Rewrite ODE → 3) Substitute → 4) **Integrate** → 5) Apply boundary V_0

Now find C_0, Using that $V(t=0)=V_0$

$$V = \exp(\frac{a}{b} - \frac{C_0}{b}\exp(-b*t))$$

$$V(0) = V_0 = \exp(\frac{a}{b} - \frac{C_0}{b}\exp(-b*0))$$

$$V_0 = \exp(\frac{a}{b} - \frac{C_0}{b})$$

$$\ln V_0 = \frac{a}{b} - C_0/b$$

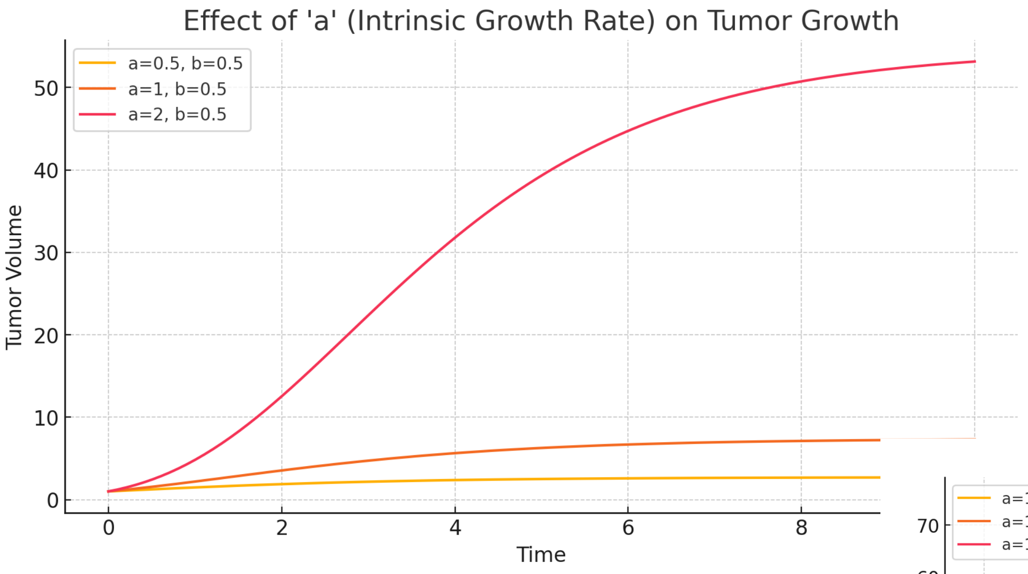
$$C_0 = a - \ln(V_0) * b$$ Insert to find solution

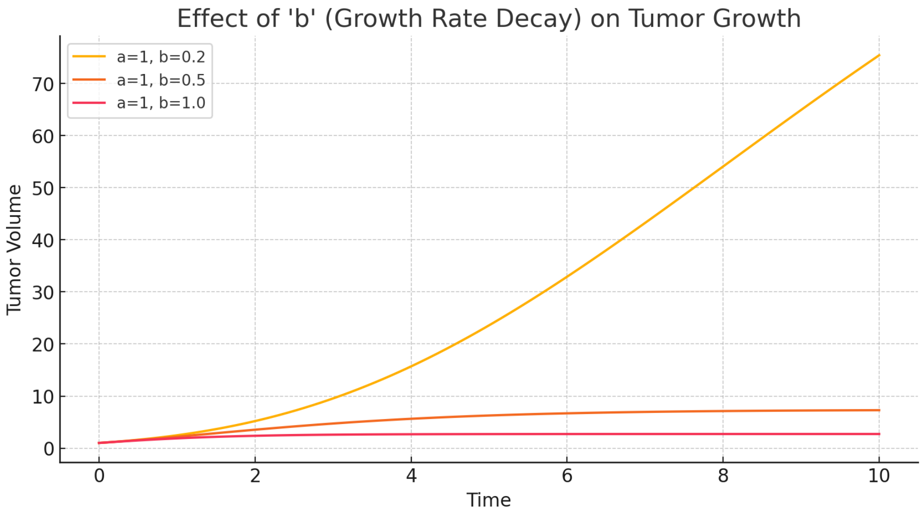
$$\boldsymbol{V(t) = exp[\frac{a}{b} - \left(\frac{a}{b} - \ln(V_0)\right) * exp(-b*t)]}$$

The term **a/b−ln(V0)** can be considered as the transformed initial volume
The expression **exp(−bt)** describes the decay over time.

model → 2) Rewrite ODE → 3) Substitute → 4) Integrate → 5) **Apply boundary $\boldsymbol{V_0}$**

Modeling Tumor Growth using the Gompertz Function





$$\frac{dV}{dt} = V * (a - b * \ln(V))$$

V0 is the initial tumor volume at t=0.

a is a growth-related parameter.

b is the rate at which the growth rate decreases over time.

$$\boldsymbol{V(t) = exp(\frac{a}{b} - (\frac{a}{b} - \ln(V_0))\, exp(-b * t))}$$

What have we seen today

- **Applications of integration**
- **Hagen–Poiseuille Law: Liquid flow through a pipe / artery**
- **Integration by substitution**
- **Modeling Tumor Growth using the Gompertz Function**